

SUMMARY

Software Engineer experienced in designing scalable backend systems, manufacturing test software, data-driven validation frameworks, automation systems, cloud-native analytics applications and automated reporting systems in C++ and python.

Proven track record of improving production yield visibility, release efficiency, and test reliability in high-volume production systems.

Strong foundation in data structure & algorithms, with additional experience in financial analytics, statistical modeling, and large-scale data systems, building OLAP models.

EXPERIENCE

Software Engineer – NAND SSD Manufacturing Test Development

Solidigm Technology, CA 📅 December 2021 – August 2023

- Built internal yield analytics platform using python, Flask and Dash supporting real-time SSD production metrics; reduced manual data analysis by ~80%.
- Developed a full-stack web application to visualize and manipulate yield on High Volume Manufacturing & Testing of SSD.
- Created a JMP based application to gather usage statistics of iPC testers.
- Designed and automated tape release validation system (BRTC), improving release cycle efficiency and minimizing production delays and standardizing the validation program release across cross-functional teams.
- Created a python based tool to perform eFAI check on NPI SSDs
- Contributed to the validation program code base (C++) for multiple NAND SSD products in NPI & HVM environments.
- Conduct experiments to root-cause debugging for new products.
- Led one of the major projects to re-enabling fused drives for customers leading to a profit in 50M.
- Mentored lab technicians and onboarded engineers on SSD test lifecycle and tooling standards.
- Conducting and assisting in technical interviews.

Product Development Engineer – SSD Manufacturing Test Development

Intel Corporation, CA 📅 July – December 2021

- Developed test program software in C++ to meet SSD specifications and high-volume manufacturing requirements.
- Contributed to the design and process improvements of the test program software-stack to modularize components of the test program to decouple dependencies of a production release.
- Analyzed and evaluated manufacturing yield data. Reported information to a small audience to correlate that the data produced by the test program meets internal targets.
- Designed elaborate debug experiments to reproduce and root cause failures seen only in the factory manufacturing environment.
- Other responsibilities included:
 - Lead group discussions to drive continuous improvement and expand our software tool capabilities.
 - Orchestrate and oversee early adoption efforts to modernize team tools to industry standards and best-practices.

Product Development Engineer Intern – SSD Manufacturing Test Development

Intel Corporation, CA 📅 January – July 2021

- Set up a data platform, one stop reporting and analytics repository for all NAND products to support NPSG SSD Manufacturing and Test Development team along with partnering teams and eliminated 90% of data processing, reports and dashboard generation done manually.
- Designed & developed:
 - Cloud native applications for statistical analysis such as Product Health Indicator, Statistical Kill Limits, Outlier Detection, Critical Parameter Comparison, Validation Checklist Report using NAND SSD Manufacturing & Test Development data.
 - Web based data reporting & interactive visualization tool using Python, Dash & Flask framework that provides real-time information of yield, test-time, drive temperature, PMIC, media status, etc.
 - Automated testing infrastructure and methods for the application(s) using Python.
 - Interactive, user-friendly, and real-time dashboards using Dash framework outlining various metrics and KPIs involving drive failure and reducing 90% of the manual data analysis work done previously.
- Created a CI/CD pipeline to auto deploy on Cloud Foundry.
- Version controlling of every application was done in Bitbucket.

Profit Optimization Intern – IOTG OM Task Force

Intel Corporation, AZ 📅 June - December 2019

- Collaborated with multiple teams – Operating Margin Task Force, IOTG-Finance, Sales & Marketing Group (SMG) – for data collection, analysis, modeling & visualization.
- Performed
 - Exploratory Data Analysis (EDA) on several terabytes of financial data to identify trends, anomalies, outliers.
 - Statistical analysis like hypothesis testing for validation and verification of data models.
- Developed and executed strategy for gathering and analyzing data, investigating data trends, anomalies, and business opportunities.
- Designed, developed, and maintained key indicators necessary to deliver analytical findings.
- Designed & developed
 - Data visualizations and real-time dashboards to drive standardization and year-round tracking of IOTG Operating Margin.
 - Executive view dashboard with SKU level margin tool for Sales & Marketing Group team to set margin goals and allow comparable margin visibility.
 - Interactive team website with one stop project management repository using SharePoint, HTML, CSS, and JavaScript.
- Extensive use of Power BI to produce operational dashboards and visualization in support of in-depth analysis.
- Achieved an accuracy of more than 85% on the data model (OLAP Cube) built for the margin tool embedded in the executive view dashboard for SMG team.

Monitoring & Visualization Engineer

HSBC HDPI, India 📅 February 2014 - 2015

- Developed a monitoring tool using Java and Google Web Toolkit to display network traffic statistics and heartbeats of a cluster of nodes.
- Created a matrix in a web portal with visualization of nodes connectivity status and network health.

Revenue Data Analyst – Modern Trade

Parle Agro, India 📅 January 2013 - 2014

- Perform statistical analysis of both primary & secondary revenue data of Modern Trade division.
- Developed a web application using Python that generates revenue reports and visualization on a daily, weekly & monthly basis when it’s fed with PO Tracking data.

Finance Analyst Intern

IDBI Capital, India 📅 April – September 2012

- Data collections, analysis, and investigate trends in stock market data of 5 years i.e., from 2007 to 2011, to study factors that contribute to market volatility.
- Developed an internal web application using HTML, CSS, and JavaScript
 - to visualize stock price fluctuations over a given duration of time.
 - automating the process to generate daily reports indicating invalid, duplicate, and corrupt date and other discrepancies which improved data quality by 80%.

Finance Data Analyst Intern

OHPC, India 📅 May – October 2011

- Developed

- web application using HTML, CSS & JavaScript for internal users to enter a transaction in system & process to finally reflect in financial reports & visualizations.
- A workflow system for classifying and tracking all financial transactions resulting in improved data accuracy.
- Workflow automation engine to validate and monitor a defined sequence of tasks in recording and processing transactions.
- Interactive visualization that provides various financial analysis information such as Liquidity, Profitability & Solvency ratios, and focusing more towards measuring the various ratios (EBIT, EBITA, Operating Profit Margin etc.) & Working Capital Analysis.

EDUCATION

M.S. in Computer Science Portland State University	 2019 – 2021
Post-Bacc in Computer Science Portland State University	 2017 – 2018
Computer Science (Transfer) Portland Community College	 2016 – 2018
MBA in Finance Regional College of Management, India	 2011 – 2013

SKILLS

Technical:

Python, R, RMarkdown, C, C++, Java, Data Structure, Algorithm, HTML, CSS, JavaScript, ReactJS, Bootstrap, SharePoint, Wireframing, Power BI, Tableau, Linux, Dash, Flask, Cloud Foundry, AWS, Google Cloud Platform (GCP), Azure, Docker, Kubernetes, GitHub, Bitbucket, MongoDB, SQL, PostgreSQL, SDLC, Object-Oriented Programming, Data Mining, Data Wrangling, Data Modelling, Data Analysis, EDA, JMP, Structural Analysis, Product Road-Mapping, JUnit Testing, JSP, Servlet, REST API, Google Web Toolkit, Android, C#, .NET, OLAP, Microsoft Office, Excel, VBA, Jupyter Notebook

Business:

Ratio Analysis, Cash Flow Management, Security Analysis, Capital Budgeting, Trading & P/L, Balance Sheet Reports, Financial Reporting.

ACADEMIC PROJECTS

Machine Learning (Python):

- **Stock Price Prediction** – implementation & performance evaluation of three Machine Learning models (i.e. Linear Regression, SVR (Support Vector Regression), LSTM (Long Short-Term Memory)) for forecasting the future stock prices. It also involves validation of the models using real datasets acquired from Google Finance, Kaggle and Yahoo Finance; and performance and accuracy evaluation of the models by comparing and analyzing it against some typical baselines.
- **Clustering** – Implemented K-Means and Fuzzy Means, unsupervised machine learning algorithms using Numpy and Matplotlib libraries, and a dataset consisting 1500 data points in 2-dimension from 3 Gaussians, with considerable overlap.
- **Spambase** – This project involves solving a classification problem in ML i.e. identifying spam/non-spam emails from Spambase datasets using linear SVMs
- **Digit Classification** – Recognized handwritten digits from MNIST Dataset by implementing perceptron learning algorithm. Trained 10 perceptron to learn and classify as a group, the handwritten digits in the MINST dataset, where each perceptron has 785 inputs consisting of 784 pixels i.e., 28x28 pixel image represented as gray scaled value (0 – 255) and one output i.e., either 0 or 1. Each perceptron’s target is one of the 10-digits (0-9) and it learns using the perceptron learning algorithm.

Data Visualization (R): Multiple data visualization projects on Museum of Modern Arts (MOMA), COVID-19, ArcGIS Maps, developing static and dynamic website using Shiny, Leaflet, ArcGIS, Tidyverse, ggplot, dplyr, RMarkdown, blogdown, Hugo, etc.

Database:

- **PostgreSQL Benchmarking (Python & SQL)** – A multi-phased project based on Wisconsin benchmarking, aspired to evaluate PostgreSQL database with assorted configuration parameter settings and optimizer options; that implicates designing and implementation of various experiments for the performance benchmarking. It involved generation and storage of variable sized (100K to 100M tuples) datasets of strings, designing and implementation of queries, working with PostgreSQL’s configuration settings, index, query optimization, joins, scans while executing the queries.
- **MongoDB (Java)** – Hands-on experience of data preprocessing & wrangling, designing schema, data modelling, storage and application design with cloud-based data management system in MongoDB Atlas, using Freeway dataset along with designing and implementation various queries in Java as well as MongoDB query language.

Frontend Web Development:

- **Marvel Comics Browser (React JS)** – Developed a user-friendly and interactive website using ReactJS to explore comics by series & super heroes of Marvel Universe.

Full-Stack Web Development:

- **Instapix (Flask, SQL, gcloud storage, Cloud Vision & Translate API)** – Adopted MVC (Model-View-Controller) architectural pattern to design and develop a Google cloud-based social networking application using Flask web framework of Python, that allow users to follow other users, share/comment/like posts/photos, and view timeline etc.
- **Bubble Tea (Flask, SQL, GCP)** – Designed and developed a Google cloud-based application to allow users to add different Bubble Tea store details (name, location, hours, contact, drinks, etc.) and give reviews.

Software Applications:

- **Web Crawler (Python, BS4)** – traverses the world wide web to create search engine index with given seed links.
- **Chat Application (Java)** – A Text-based messaging application using Java swing for designing GUI, HTML for aligned display of messages, Java socket for client-server communication, MongoDB to persist user’s information.
- **Phone Bill (Java, GWT)** – web application that maintains the phone call record details. Features implemented are: display a README manual describing how the program works, adding a new call record, display details of all calls or calls recorded within a certain period of time.
- **Twitter (Java)** – An application which emulates Twitter. Given a set of tweets as input, it extracts information such as date and time, user, tags, keywords, etc. and categorizes the tweets based on it.

Data Structures & Algorithm (Java & C++):

- **String Matching** – Given a text, find the occurrences of a pattern in it and return the locations of match. Implemented Brute force, Boyer Moore, KMP, Shift-OR, Rabin-Karp, in C++ to performance benchmark multiple string-matching algorithms given randomly generated patterns & Homo_sapiens DNA sequence datasets of variable sizes, to find the one with the best and the worst-case run-time and space complexity.
- **Sorting Algorithms** – performance evaluation of various sorting algorithms using randomly generated datasets.
- **Graph Algorithm:** implemented Kruskal’s and Prim’s method of finding the Minimum Spanning Tree on random graphs, using union find data structure.
- **Shunting Yard** – Dijkstra’s shunting yard algorithm using stack to evaluate logical expressions with AND, OR, NOT, etc. based on their order of precedence.
- **Multiply linked list** – implemented linked list with three separate link chains to keep the data entries unique and sorted by three different fields.
- **AVL Tree** – set container like in STL using AVL tree that supports union, intersection, cross product, and power set operations.

Operating Systems:

- **XV6 (C, UNIX)** – Implemented **System calls:** a new system calls for date and time. **Process:** system calls to support and obtain process ownership, set and extract UID, GID, PID, track amount of time a process uses CPU. **Shell:** create new user commands (‘ps’ and ‘ctrl+p’ command), extract process detailed information. Scheduling: implemented MLFQ scheduler. **File system:** chmod, chown, chgrp system calls to set and modify file permissions that persist across xv6 sessions. Modified ‘exec’ and ‘ls’ command to enforce and display new file permissions respectively.
- **Buddy System (C)** – module that imitate malloc and free to manage a block of memory (e.g. heap)

REFERENCES

Available on request.

